



COURSE OUTLINES

BACHELOR OF SCIENCE IN MATHEMATICS

NATIONAL UNIVERSITY OF PAKISTAN



Course Outlines for
Mathematics
BS Program

2025-2029

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Guiding Principles

The curriculum serves as a comprehensive blueprint for the teaching-learning process that students must navigate through to achieve specific academic objectives. This encompasses clearly defined objectives, a scheme of studies and learning outcomes. Knowledge rapidly evolves and new fields emerge so it is crucial to develop such curricula as are current, relevant, and impactful.

As mandated by its Charter through Section 31(2) (i), National University of Pakistan has developed curricula for various disciplines of BS Program. The criteria, schemes of studies and course outlines of HEC set the benchmark for development of schemes of studies and course outlines at National University of Pakistan. The subject committees, appointed for development of curricula, comprised of subject matter experts and researchers. In response to the evolving needs of the 21st century, National University of Pakistan, in collaboration with HEC, undertook the task of developing robust standards of curricula for BS Program.

It is hoped that this document, developed by dedicated subject experts from across the country, will not only fulfill our national educational and economic requirements but also elevate the competency levels of our graduates.

With support of affiliated colleges in implementation of these curricular standards, National University of Pakistan envisions a future where its graduates are at the forefront of scientific discovery and technological innovation, driving progress and contributing to the betterment of the society both nationally and internationally.

BS Mathematics

Program Description

The Bachelor of Science in Mathematics at NUP is designed to cultivate a deep understanding of mathematical principles and their applications across diverse fields. Recognizing the pivotal role of mathematics in driving innovation in areas such as economics, business administration, and natural sciences, computation sciences, the program offers a comprehensive curriculum that equips students with the necessary skills to excel in these domains.

Over four years, students will delve into core mathematical concepts while specializing in areas like linear algebra, calculus, mechanics, numerical analysis with computer applications, differential geometry, mathematical statistics, mathematical modeling, computer programming languages, discrete math, graph theory, and operations research. This robust foundation will prepare graduates for successful careers in a wide range of industries, including banking and the economic sector, industrial research organizations, statistical organizations, software houses, nuclear sectors, operational research firms, engineering, data science, and risk analysis.

Program Structure

Minimum Credit Hours	136
General Education Courses	34 credit hours (15 courses)
Disciplinary Courses	75 credit hours (24 courses)
Interdisciplinary Courses	21 credit hours (7 courses)
Internship / Field work	3 credit hours (6 to 8 weeks)
Capstone Project	3 credit hours
Program Duration	Minimum: 4 Years Maximum: 6 Years The maximum limit is further extendable to another year in extra-ordinary circumstances subject to approval of the Rector.
Semester Duration	18 weeks
Course Load (per semester)	15-18 credit hours for regular semesters (21 credit hours for remedial/deficiency/failure/repetition courses only)
3 Credit Hours (Theory)	3 classes (1 hour each) OR 2 classes (1.5 hour each) OR 1 class (3 hours) per week throughout the semester.
1 Credit Hours (Lab / Field Work)	1 credit hour in laboratory or practical work / project requires lab contact of three hours per week throughout the semester.

Scheme of Studies

Sem	Course Code	Course Title	Cluster/ Domain	Credit Hours (Theory + Lab)
1 st	GEN-101	Functional English	Gen Ed	3
	GEN-102	Quantitative Reasoning-I	Gen Ed	3
	GEN-103	Application of Information and Communication Technologies	Gen Ed	3(2+1)
	GEN-125	Understanding of Quran-I	Gen Ed	(0,1)
	IDP-201	Introduction to Mechanics	Interdisciplinary	3
	MATH-101	Calculus-I	Disciplinary	4
	Total Credit Hours			17
2 nd	GEN-104	Expository Writing	Gen Ed	3
	GEN-105	Quantitative Reasoning-II	Gen Ed	3
	GEN-106	*Islamic Studies	Gen Ed	2
	MATH-111	Calculus-II	Disciplinary	4
	MATH-112	Elementary Set Theory	Disciplinary	3
	STAT-112	Statistics	Interdisciplinary	3
	Total Credit Hours			18
3 rd	GEN-107	Environmental Science	Gen Ed	3
	GEN-110	Ideology and Constitution of Pakistan	Gen Ed	2
	GEN-113	Islamic History and Culture	Gen Ed	2
	GEN-126	**Understanding of Quran-II	Gen Ed	(0,1)
	MATH-201	Calculus-III	Disciplinary	4
	MATH-202	Discrete Mathematics	Disciplinary	3
	IDP-301	Software Packages	Interdisciplinary	3(2+1)
	Total Credit Hours			18
4 th	GEN-111	Entrepreneurship	Gen Ed	2
	GEN-112	Civics and Community Engagement	Gen Ed	2
	GEN-109	Sociology	Gen Ed	2
	GEN-124	Pakistan Studies	Gen Ed	2
	MATH-212	Linear Algebra	Disciplinary	3
	MATH-213	Vector and Tensor	Disciplinary	3
	IDP-211	Introduction to Machine Learning	Interdisciplinary	3(2+1)

	Total Credit Hours			17
5th	MATH-301	Real Analysis-I	Disciplinary	3
	MATH-302	Ordinary Differential Equations-I	Disciplinary	3
	MATH-303	Group Theory	Disciplinary	3
	MATH-304	Complex Analysis	Disciplinary	3
	MATH-305	Topology	Disciplinary	3
	IDP-302	Classical Mechanics	Interdisciplinary	3
	Total Credit Hours			18
6th	MATH-311	Real Analysis-II	Disciplinary	3
	MATH-312	Ordinary Differential Equations-II	Disciplinary	3
	MATH-313	Differential Geometry	Disciplinary	3
	MATH-314	Classical Mechanics	Disciplinary	3
	MATH-315	Functional Analysis	Disciplinary	3
	IDP-311	Computer Programming	Interdisciplinary	3(2+1)
	Total Credit Hours			18
7th	MATH-401	Numerical Analysis-I	Disciplinary	3
	MATH-402	Partial Differential Equations	Disciplinary	3
	-	Elective Course-I	Disciplinary	3
	-	Elective Course-II	Disciplinary	3
	IDP-401	Scientific Writing & Research Methods	Interdisciplinary	3
	Total Credit Hours			15
8th	MATH-403	Numerical Analysis-II	Disciplinary	3
	INT-400	Field Experience	Field Work	3
	CAP-425	Capstone Project	Capstone Project	3
	-	Elective Course-III	Disciplinary	3
	-	Elective Course-IV	Disciplinary	3
	Total Credit Hours			15
	Total Credit Hours of the Program			136

***Non-Muslims may opt 'GEN-123 Ethics' as a substitute.**

****Non-Muslims may opt 'GEN-130 Study of World Religions' as a substitute of Understanding of Quran in Semester-IV.**

List of Elective Courses for 7th (any One) & 8th Semesters (any Two)

Course Codes	Course Title	Cr. Hrs.	Semester
MATH-404	Measure Theory and Lebesgue Integration	3+0	7 th
MATH-407	Rings and Vector Space	3+0	7 th
MATH-406	Theory of Modules	3+0	7 th
MATH-407	Theory of Group Action	3+0	7 th
MATH-408	Analytical Dynamics	3+0	7 th
MATH-409	Introduction to Quantum Mechanics	3+0	7 th
MATH-410	Optimization Theory	3+0	7 th
MATH-411	Mathematical Statistics-I	3+0	7 th
MATH-412	Special Theory of Relativity	3+0	7 th
MATH-413	Fluid Mechanics-I	3+0	7 th
MATH-414	Operations Research-I	3+0	7 th
MATH-415	Fortran Programming	3+0	7 th
MATH-416	Advanced Number Theory	3+0	7 th
MATH-417	Special Functions	3+0	7 th
MATH-418	Theory of Splines	3+0	7 th
MATH-419	Algebraic Systems and Coding Theory	3+0	8 th
MATH-420	Methods of Mathematical Physics	3+0	8 th
MATH-421	Advanced Topology	3+0	8 th
MATH-422	Electromagnetism	3+0	8 th
MATH-423	Calculus of Variation	3+0	8 th
MATH-424	Advanced Group Theory	3+0	8 th
MATH-425	Mathematical Modeling	3+0	8 th
MATH-426	Operational Research-II	3+0	8 th
MATH-427	Fluid Mechanics-II	3+0	8 th
MATH-428	Graph Theory	3+0	8 th
MATH-429	Cryptography	3+0	8 th
MATH-430	Mathematical Statistics-II	3+0	8 th
MATH-431	Computer Applications	3+0	8 th
MATH-432	Heat Transfer	3+0	8 th
MATH-433	Riemannian Geometry	3+0	8 th
MATH-434	Integral Equations	3+0	8 th
MATH-435	Rings and Fields	3+0	8 th

Semester-I Course Outlines

Functional English
GEN-101

Program-Discipline	BS - Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Functional English**Course Code:** GEN-101**Course Description**

The course covers the fundamental concepts of English composition and comprehension. The course is designed in such a way that students can use this knowledge to further enhance their language skills in English. The course aims at enhancing students' skill and competence in communicating their ideas in writing and speaking in English language. It primarily focuses on four areas of language i.e. listening, reading, the conventions of written English, speaking and building build-up vocabulary to help students achieve proficiency in language use.

Course Learning Outcomes (CLOs)

Upon successful completion of the course, the student will be able to:

CLO No.	CLO Description
CLO-1	To <u>write</u> coherently and cohesively. To write and reply applications/ letters/ emails. To construct grammatically correct oral and written sentences. To tell and write stories/ essays.
CLO-2	To <u>pronounce</u> words in English language correctly. To <u>read</u> and interpret text written in English language. To <u>listen</u> and decipher spoken forms of English language.
CLO-3	To <u>improve</u> writing, presentation, listening and communication skills.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3×16) 48 contact hours in the semester. In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	- Basics of Grammar - Parts of Speech
2	Vocabulary Building (Use the Provided Booklet) - Synonyms - Antonyms - Homonyms
3	Vocabulary Building (Use the Provided Booklet) - One word Substitution - Words often confused (Pairs of Words)
4	Vocabulary Building (Use the Provided Booklet) Verbal Phrases and Idiomatic Phrases (including Phrasal Verbs)
5	Sentence Structure (simple, compound, complex and compound-complex)
6	Tenses
7	Elimination of Common Errors (Correction) Subject-Verb Agreement, Noun
8	Elimination of Common Errors (Correction) Adjective, Verb, Adverb
9	MID TERM EXAMS
10	Listening - Quiz on audios - Quiz on videos
11	Comprehension/ Reading Skill - Critical Reading - SQ4R - Reading of passages and comprehension exercises (Level Advance)
12	Public speaking - Informal communication (small talk, conversational skills) - Public speaking (overcoming stage fright, voice modulation and body language)

13	Controlled and Guided Writing • Pre writing (Brainstorming, mind-mapping, planning, information gathering, preparing to write), writing, search for topic sentences • Developing a theme, following up ideas and arguments, outline plans etc.
14	Paragraph Writing • Topics at the discretion of the teacher
15	Precis Writing • Rules of précis writing • Practice précis.
16	Essay Writing • Types of writing: descriptive, narrative expository, argumentative
17	Essay Writing • Expository, argumentative
18	END SEMESTER EXAMS

Recommended Books

1. College Writing Skills with Readings, by John Langan, McGraw-Hill, 5th Edition.
2. High School English Grammar & Composition by Wren & Martin
3. Practical English Grammar by A. J. Thomson and A. V. Martinet

Quantitative Reasoning-I
GEN-102

Program-Discipline	BS - Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Quantitative Reasoning-I**Course Code:** GEN-102**Course Description**

The course objectives to provide students with an understanding of the basic Mathematical and Financial Techniques used in Business Enterprises or Accountancy Profession through application of techniques rather than its theoretical aspect. To equip students with Mathematical Skills and Knowledge that can be applied to solve Financial Problems in field of Accountancy, Enhancement of students' competency in application of mathematical linear & quadratic, Equations in solving business management problems. Understanding the concepts of matrices and its applications. Derivatives and its role in business and economic approach

Learning Outcomes

Upon the successful completion of this course, students should have learnt:

- Mathematical Function
- Building and solving linear and quadratic equations
- Types of functions
- Matrices and its applications
- Determinants and its applications
- Derivatives & its business applications

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3×16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	- Cartesian Product (Basic Examples only) - Relations, Types of relations (Reflexive, Symmetric, Transitive and Equivalence relation)
2	- Definition: Function, Domain, Co-domain, Range - Types (one one, onto, Into, bijective). - Identifying type of a given function.
3	Set Operations (subsets, Power Set, Set Notations, Union and intersection of sets), Compliments
4	Logic, Simple Statement, Compound Statement, negation, Conjunction, Disjunction, Conditional, biconditional, Tautology, absurdity.
5	- Translating from Symbols to English and vice versa., - Constructing Truth Table, logical Equivalence, De Morgan's Laws.
6	- Introduction of matrices, Types of matrices (row, column, square, identity, zero, diagonal) - Addition and subtraction of matrices, Multiplication of Matrices (2×2 & 3×3) .
7	- Determinant and inverses, Minor and cofactor up to 3×3 matrices. - Inverse of a matrix by using adjoint method (up to 3×3)
8	- To transform a Matrix of order 3×3 to row echelon form - To determine if a given matrix is in row echelon form or not.
9	MID TERM EXAMS
10	- Definition of Equation, Linear equations containing one or more variables and operations on these equations, - Simultaneous Equations in two variables and methods of solving such problems (By elimination, By substitution)
11	Solution of System of equations containing three variables by Matrix Inversion method, Cramer's rule.
12	- To transform a Matrix of order 3×3 to reduced row echelon form - Linear equations in three variables and their solution by

	Gaussian Elimination method, Gauss Jordan elimination method.
13	Slope (Positive, Negative, Zero, Undefined slopes) and its interpretation, X and Y intercepts and their interpretation.
14	- Finding slope, a) If inclination is given b) If two points are given.
15	Formation of Equation of straight-line using point slope form, slope intercept form, Two-point form.
16	Conversion of straight-line equation into point slope form, slope intercept form, two-point form.
17	Parallel and perpendicular lines.
18	END SEMESTER EXAMS

Recommended Books

1. Business Mathematics by Frank.S. Budnick Latest Edition.
2. Business Mathematics by Gary Clendenen and Stanley A. Salman 13th edition.

Reference Book

1. Text book of 1st Year & 2nd Year of SNC of Punjab Curriculum and Textbook board

**Application of Information and
Communication Technologies
GEN-103**

Program-Discipline	BS – Mathematics
Credit Hours	3 (2+1)
Duration	18 Weeks

Course Title: Application of Information and Communication Technologies

Course Code: GEN-103

Course Description
This introductory course in Computer Science provides a comprehensive exploration of fundamental computing concepts and practical applications. The course is designed for beginners and covers the history of computing, computer hardware and software, computer fundamentals, web programming, and the utilization of popular productivity tools. Theoretical sessions span various aspects of computing, beginning with the definition of computers and their historical evolution. This course cover major topics related to fundamentals of computer, operating system, data communication, networking, computer security and World Wide Web. Practical insights into HTML and web page creation will be covered to enhance the students' understanding of applied technologies. Laboratory sessions focus on hands-on applications of computing skills. Students will engage in practical exercises covering email communication, operating system tasks, Microsoft Office suite proficiency (PowerPoint, Word, Excel), and introductory HTML for web page creation. The lab sessions aim to strengthen theoretical concepts through practical application and develop essential skills for professional computing.

Course Learning Outcomes (CLOs)

Upon successful completion of the course, the student will be able to:

CLO No.	CLO Description
CLO-1	<u>Understand</u> basics of computing technology i.e. Computer Hardware and Software, Basic Operations of computers, Basic OS Concepts.
CLO-2	<u>Understand</u> and can-do number systems conversions and arithmetic.
CLO-3	Have <u>knowledge</u> of network components, communication medium, Computer security and privacy.
CLO-4	<u>Analyze</u> how various applications can be used to solve different problem. Students will be able to <u>create</u> professional documents, spread Sheets and presentations using productivity tools such as MS Word, MS Excel, MS PowerPoint.
CLO-5	<u>Identify and Explore</u> valuable resources for education and research and <u>synthesize</u> information from multiple sources to achieve objectives. Students will be able to <u>use</u> email for professional communication. Students will be able to <u>create</u> professional Profile/CV.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

Practical / Lab Courses

	Exam	Weightage
(a)	End-Semester Exam	30
(b)	Mid-Semester Exam	20
(c)	Experiment/Lab Reports / Viva Voce / Project / PBL (Problem Based Learning)	50
	Total	100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline (Theory)

Week	Contents
1	Definition of Computer, Computer Processing System (Input, Process, Output), Difference between data and information, Role of Computers in various fields (Education, Business, Media, Manufacturing), History & Generations of Computers with Characteristics of each generation.
2	Types of Computers (Analog, Digital, Hybrid), Types of Digital Computer (Super, Mainframe, Mini and Micro Computer), Introduction to CBIS (Computer Based Information System), Methods of Input and Processing. Basic Operations of a computer (Input, Process, Storage, Output), Computer Elements, Processor, Memory, Hardware, Software.
3	Computer Hardware: Input devices (Keyboard, mouse etc), Output Devices (Speakers, Monitors and its Types, Printers and its Types, Plotters and its types).
4	Storage Units, Primary and Secondary Memories and their types, RAM and its Types, Cache, Hard Disks, RAID, Optical Disk Storages (DVD, CD ROM), flash memory storage systems (memory cards and readers, USB flash drives) Other types of storage systems (network and cloud storage system, smart cards)
5	Types of Computer Software: System Software and Types, Operating System, Device Drivers, Utility Software (backup and recovery utilities), Language Processors (Assembler, Compiler and Interpreter), Importance, Uses, Limitations and Types of Application Software (Productivity Software, Business Software, Entertainment Software, Educational Software), Internet Applications (Web Applications, Cloud Computing Applications, Social Media Network Applications), Others (Licensed Software, Open source Software, Shareware/Trialware, Freeware).

6	Software Development life Cycle (SDLC), Importance of SDLC, SDLC Phases/Steps, Personnel involved in SDLC (Project Manager, System Analyst, Programmer, Software Tester, Customer or End User).
7	Introduction to Operating System. Functions of Operating System, Types of Operating System interfaces (Command Line Interface, Menu Driven Interface, and Graphical User Interface), Examples of CLI (DOS, UNIX) and GUI (MacOS, Linux, and Windows).
8	Introduction to Mobile OS (Android and IOS)
9	MID TERM EXAMS
10	Data Communications, Data Communication Model, Data Transmission, Digital and Analog Transmission, Modems, Asynchronous and Synchronous Transmission, Modes of Communication: Simplex. Half Duplex, Full Duplex Transmission,
11	Communications Media: Guided and Unguided Medias (Cables, Wireless), Protocols (http, TCP/IP, FTP, POP3), Network Topologies (Star, Bus, Ring, Mesh), LAN, MAN, WAN, PAN.
12	Security and privacy: Malware and its Types (Virus, Worm, Adware, Spyware), Computer Virus and its Forms., Authentication and Authorization, Symptoms of Malware, How Malware Spreads, Precautions to protect computers.
13	Number Systems and Conversions: Binary, Octal, Decimal, Hexa-Decimal.
14	Internet, A Brief History, Internet Services provider, DNS, web Hosting.
15	Introduction to World Wide Web, Terms related to WWW (Web Page, Website, Web Browser, Web Server, Uniform Resource Locator and its different parts, Search Engine, Home Page).
16	Types of Websites (Web portals, News Websites, Informational Websites, Educational Websites, Personal Websites, Business Websites, Entertainment Websites)
17	Ethical and responsible use of ICT, Digital citizenship, Cybercrime and its types, Intellectual property rights.
18	END SEMESTER EXAMS

Week Wise Course Outline (Lab)

Week	Contents
1	Email: Creating an Email Account, Create and Send Email, use to various folders (Inbox, Drafts, Spam, and Sent folder etc.), Use of "To", "Subject", "Attach files button", "Send Button", "Composing Area", "CC" and "BCC". Working with Received Emails (Reading an Email, Downloading Attachments, replying to an Email), Sign out and Email Account. LMS (Google Classroom)
2	Operating System: Create file and folders, Move or delete file and folders, Basic Shortcut keys (Cut, Copy, Paste, Delete, select all etc), Windows Task Manager (ALT+CTRL+DEL), Control Panel Overview (System and Security, User Accounts, Appearance and Personalization, Network and Internet, Clock, Language and Region, Uninstall a Program), Command Prompt Overview and Commands (dir, cd, md, rd).
3	MS Power Point: Interface (Title Bar, Quick Access Toolbar, Ribbon, Horizontal and Vertical Scroll Bars, Status Bar, Zoom Slider, Notes Pane etc), Creating a Power Point Presentation, apply colors, fonts and effects, changing fill style and background, Selection of Slide layout, Add and Remove Slides, Add Text in Slides, running a Presentation, Saving Presentation, Difference Between Save and Save As.
4	MS Power Point: Formatting Text, Addition of Clipart, Addition of a Drawing, Adding Animations to the Presentation, Adding Pictures, Videos or Sounds in Presentation. Applying Slide Transition,
5	MS Word: Interface (Title Bar, Quick Access Toolbar, Ribbon, Horizontal and Vertical Scroll Bars, Status Bar), Creating a New Microsoft Word Document, opening a Document, Saving and Printing a Document, Inserting and Editing Text, Selecting Text, Copy and Moving Text (Ctrl + X, Ctrl + C, Ctrl + V), Deleting text. Inserting Page Borders, Inserting Table of Contents.

6	MS Word: Inserting Symbol, Inserting Equation, Formatting Text (Changing the Font Type and Size of Text, Changing Font Styles and Effects, Changing Case, Use of Bold, Italics, Underline, Strikethrough, Subscript, Superscript, Text Highlight Color and Font Color), Formatting Paragraph (Changing Paragraph Alignment: Align left Align Centre, Align Right, Justify ,Indenting Paragraph: Increase and decrease indent, Changing Spacing between Paragraphs and Lines), Formatting Page (Document Theme, Page Margins, Page Orientation: Portrait and Landscape, Page Background: Watermark, Page Color and Page Borders). Use of Find and Replace.
7	MS Word: Inserting Header, footer and Page numbers, Inserting Cover page, Inserting Page and Section Breaks, Inserting and Positioning Picture, Inserting Word Art
8	MS Word: Creating and Formatting table, Adding and Deleting Rows or Columns, resizing rows and columns, Adding or Changing Border Lines and Shading, Merging and Splitting Cells, Changing Margins, Using Hyperlink, make regular text as hyperlink, Remove Hyperlink.
9	MID TERM EXAMS
10	MS Excel: Opening the Microsoft Excel Program, Interface (Title Bar, Quick Access Toolbar, Cell, Row, Column, Formula Bar, Cell Address, Active Cell Address, Worksheet Window), Filling Columns and Rows, Auto fill and its options, Fill Series, Use of formulas (total, percentage, average etc), Merge and Split Cells, Insert Border on table, Wrapping Text, Use of Built in Functions (sum, product, average, max, min, power, sqrt, count).
11	MS Excel: Cell Addressing and its types (Relative Cell Addressing, Absolute Cell Addressing), Paste Special and its various options.
12	MS Excel: Resizing Rows and Columns, Insert New Rows and Columns in between, Delete Rows and Columns, Rename Sheets and Add Sheets in Excel Workbook. Graphical Representation of Data (i.e. Creating Column Chart).

13	MS Excel: Manipulating and Formatting Data (Filtering Data, Data Validation, protecting a Worksheet, Unlocking Cells, Conditional Formatting).
14	MSPaint, Elementary Video editing
15	Video Conferencing (Zoom, Google Meet).
16	Google Forms (creating data collection forms, surveys etc).
17	Development of professional CV & linked in profile (Semester Project)
18	END SEMESTER EXAMS

Recommended Books

1. Charles S. Parker, Understanding Computers: Today and Tomorrow, Course Technology, 25 Thomson Place, Boston, Massachusetts 02210, USA
2. Livesley, Robert Kenneth. An introduction to automatic digital computers. Cambridge University Press, 2017.
3. Zawacki-Richter, Olaf, and Colin Latchem. "Exploring four decades of research in Computers & Education." Computers & Education 122 (2018): 136-152.
4. Sinha, Pradeep K., and Priti Sinha. Computer fundamentals. BPB publications, 2010.
5. Alan Evans, Kendall Martin, Mary Anne Poatsy. Technology in Action. Pearson Education Inc. © 2024, ISBN 978-0-13-793278-8.

Understanding of Quran-I**GEN-125**

Program-Discipline	BS – Mathematics
Credit Hours	(0,1)
Duration	18 Weeks

Course Title: Understanding of Quran-I

Course Code: GEN-125

Course Description
This required course focuses on writing clearly and cogently, overcoming prevalent errors in language, mastering rules in grammar and syntax, and developing skills in organizing and structuring different forms of writing. The course consists of two major parts: the reading section-focuses on recognizing a topic sentence, skimming, scanning, use of cohesive devices, identifying facts and opinions, guess meanings of unfamiliar words etc. The writing section deals with the knowledge and the use of various grammatical components and essay writing skills which would enable students to write effective and advanced expository writing.

Course Learning Outcomes (CLOs)

Upon successful completion of the course, the student will be able to:

- Develop the ability to understand basic words of the Quran, phrases and sentences that do not contain verbs (unit 1 to 5 of Muallim ul Quran Book) and then sentences having present tense (first half of unit 6 of Muallim ul Quran Book).
- Acquire a strong foundation for understanding long verses of the Quran with clarity.
- Comprehend Quranic vocabulary, particles (operative & non-operative particles), compounds (Adjective & Possessive compound), pronouns (singular & plural) and types of plural through hundreds of Quranic sentences.
- Recognize and understand different styles of Quranic sentences, including nominal sentence, emphatic sentence, double emphatic sentence, negative sentence, interrogative sentence, oath –based sentences.
- Strengthen understanding of fundamental Quranic linguistic styles, expressions and idioms.
- Understand at least 30to 40%of each page of the holy Quran.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

Practical / Lab Courses

	Exam	Weightage
(a)	End-Semester Exam	30
(b)	Mid-Semester Exam	20
(c)	Experiment/Lab Reports / Viva Voce / Project / PBL (Problem Based Learning)	50
	Total	100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester. In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. 3x16 = 48 contact hours in the semester.

Week Wise Course Outline

Week	Lessons	Assignments/Home Task	Linguistic Rules
1	1-6	Writing the meaning of Quranic words Lesson 1-8	Proper Noun Masculine & Feminine
	9-14	Writing the meaning of Quranic words 9-14	Two kinds of plural Concept of (و) "And" Common Noun
2	15-17	Writing the meaning of Quranic Words, phrases & Translation of Sentences 15-17	Demonstrative Noun (This & That for Masculine (هنا - هذه) Demonstrative Noun (This & That for Feminine) (ذلك - تالك)
	18-19 & Revision (Unit 1)	Writing the meaning of Quranic words, Phrases & Translation of sentences 17-19 QUIZ	Laam for emphasis (لام التأكيد) Superlative Degree like أكبر Revision of all Quranic Sentences
3	1-3	Writing the meaning of Quranic words, phrases & translation of Sentences 1-3	Emphatic Particle (إِنَّ) Preposition "for" (في) (اللام)
	4-6	Writing the meaning of Quranic words, phrases & translation of Sentences 4-6	Preposition (على - من - إلى)
4	7-9	Writing the meaning of Quranic words & translation of Sentences 7-9	Preposition (بِ) Absolute Negation Particle Exceptive Particle (لا النافية للجنس) (ما النافية - إلا)
	10-13 & Revision (Unit 2)	Writing the meaning of Quranic words, phrases & translation of Sentences 10-13 QUIZ	Subordinating Conjunction (أَنَّ), Was (كان), Vocative Particle (حرف النداء)
5	1-2	Writing the meaning of Quranic phrases 1-2	Quranic Adjective Compounds (صفت وموصوف)

	3-5	Writing the meaning of Quranic words, phrases & translation of Sentences 3-5	Quranic Possessive Construction (مضاف ومضاف اليه)
6	6-7	Writing the meaning of Quranic phrases & Translation of Sentences 8-10	Quranic Possessive Construction (مضاف ومضاف اليه)
	8-10 & Revision (Unit 3)	Writing the meaning of Quranic words, phrases & translation of Sentences 3-5 QUIZ	Active Participle (اسم الفاعل), Passive Participle (اسم المفعول), Dual (مثنى)
7	1-2	Writing the meaning of Quranic phrases & Translation of Sentences 1-2	Personal Pronoun He (هو المنفصل) Possessive Pronoun His (هو المتصل)
	3-4	Writing the meaning of Quranic words, phrases & translation of Sentences 3-4	Possessive Pronouns with Prepositions like في بيته Pronoun "His" with prepositions like له، منه، فيه
8	5-8	Writing the meaning of Quranic Sentences 5-8	Personal Pronoun You (له - منه - فيه) Possessive Pronoun Your (أنت المنفصل) Possessive Pronouns with Prepositions like في بيتك Pronoun "your" with prepositions Like، لك، منك، فيك،.
9	MID TERM EXAMS		
10	9-11	Writing the meaning of Quranic phrases & sentences 9-11	Personal Pronoun She (هي المنفصل) Possessive Pronoun Her (ي المتصل) Possessive Pronoun with prepositions Like في بيتها Pronoun "Her" with prepositions like لها،
	12-14	Writing the meaning of Quranic phrases & sentences 12-14	Personal Pronoun (أنا المنفصل) Possessive Pronoun Her ي المتصل

11	15-16	Writing the meaning of Quranic phrases & sentences 15-16	Possessive Pronoun with Prepositions like Pronoun "My" with prepositions like لي
	17 & Revision (Unit 4)	Revision of all Quranic sentences of Unit 4 QUIZ	Adverb (حال)
12	1-2	Writing the meaning of Quranic phrases & sentences 1-2	Masculine Plural جمع المذكر السالم وجمع المذكر السالم المسبوق بحرف الجر
	3-4	Writing the meaning of Quranic phrases & sentences 3-4	Possessive Construction with Plurals جمع المذكر السالم المسبوق بالإضافة
13	5-6	Writing the meaning of Quranic phrases, sentences & verses 5-6	Personal Pronoun They (هم المنفصل) Personal Pronoun Their (هم المتصل)
	7-8	Writing the meaning of Quranic phrases, sentences & verses 7-8	Possessive Pronoun with Prepositions like في بيتهم Pronoun "Their" with prepositions like لهم
14	9-11	Writing the meaning of Quranic phrases, sentences & verses 9-11	Personal Pronoun You (أنت المنفصل) Personal Pronoun Your (كم المتصل) Personal Pronoun with Prepositions في يكم
	12-14	Writing the meaning of Quranic phrases & sentences & verses 12-14	Pronoun "Your,, with prepositions like لكم Personal Pronoun We (نحن المنفصل) Possessive Pronoun Our like (نا المتصل)
15	15-16	Writing the meaning of Quranic sentences & verses 15-16	Possessive Pronouns with Prepositions like في بيتنا Pronoun "Our" with prepositions like لنا
	17-18	Writing the meaning of Quranic sentences & Verses 17-18	Demonstrative Pronoun: These, Those (هؤلاء – أولئك)

16	19-20	Writing the meaning of Quranic sentences & Verses 19-20	ما / إلا، إن / إلا، إنما، ليس، ما، (/إم، أن، بل، كان،
	21-22	Writing the meaning of Quranic sentences & Verses 21-22	(ألا، أليس، اليوم، يومئذ، سبحان، ما بينها، قل، إذن، بش، نعم، كلا، ما أدراك، حسب، أعلم ب، مصير، مرجع، دينا (تتميز)
17	23	Translation of Quranic Sentences & Verses 23	(ألا، أليس، اليوم، يومئذ، سبحان، ما بينها، قل، إذن، بش، نعم، كلا، ما أدراك، حسب، أعلم ب، مصير، مرجع، دينا (تتميز)
	Revision (Unit 5)	QUIZ	Revision of all Quranic Sentences
18	END SEMESTER EXAM		

Recommended Book

1. Muallim ul Quran (Volume1&2) by Dr. Ubaid ur Rahman

Other material, content and books:

1. Paper book: All volumes are available in printed book form.
2. Tutorial videos: Teaching video of each lesson available on YouTube.
3. Confirmation Videos: A complete series of confirmation videos of all lessons is available, in which the student can confirm his answers.
4. A flipbook: A flipbook edition is also accessible.
5. Helping material: Helping material for the teachers like quizzes, question papers and images is available on website.

Introduction to Mechanics**IDP-201**

Program-Discipline	BS – Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Introduction to Mechanics

Course Code: IDP - 201

Course Description

This course provides a comprehensive overview of vector analysis. It gives basic understanding of laws of motion and their applications in daily life. This course also covers basic topics of mechanics like particle dynamics, collisions, system of particles, work and kinetic energy. Students will also learn about gravitation, rotational motion and angular momentum. They will learn mathematical concept and expressions of various physical parameters used in mechanics and will apply the acquired knowledge to solve problems.

Learning Outcomes

By the end of the course students would be able to:

- Understand laws and forms of motion and their applications in daily life.
- Demonstrate proficiency in particle dynamics, work, energy, system of particles, collisions and the concept of center of mass.
- Acquire knowledge and understanding of rotational dynamics and angular momentum.
- Master the fundamentals of laws of gravitation and planetary motion and applications of Kepler's laws.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester. In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	Basic concept of units, vectors and scalars, Addition of vectors by graphical and component method. Multiplication of vectors, vectors in three dimensions and fundamental operations.
2	Direction cosine, Vector and scalar Triple products.
3	Concept of Gradient, Gradient of a scalar, Divergence and Curl of a vector, Physical significance of each type, Divergence Theorem, Stoke's theorem.
4	Newton's laws of motion and their applications, Equation of motion, Projectile motion, Drag force & motion of projectiles.
5	Uniform circular motion, forces laws, tension & normal force, the dynamics of uniform circular motion
6	Collisions, linear Momentum, Impulse, Conservation of Momentum, two body collisions, one dimensional collision in center of mass reference frame.
7	Motion of a system of particles, two particle system, many particle systems,
8	Revision
9	MID TERM EXAMS
10	Work done by constant force, Work done by variable force, work done by spring force, The Work-energy Theorem, Inelastic collision.
11	Newton law of Universal gravitation, Gravitational potential energy, Escape velocity.

12	• Introduction of motion of planets and Kepler's laws, rotational quantities as vectors, rotation with constant angular acceleration, Relationship between linear and angular variables.
13	• Rotational kinetic energy, Torque, Rotational inertia and Newton's second law.
14	• Angular momentum of a particle, Angular momentum of a system of particles
15	• Conservation of angular momentum
16	• Application of law of conservation of angular momentum.
17	Revision
18	END SEMESTER EXAMS

Recommended Books

1. R. Resnick, D. Halliday, Krane. "Physics" Volume 1, 5th edition.
2. D. Holliday R Resnick and J Walker. "Fundamental of Physics, " John Willey & Sons, 9th ed. 2010.
3. R.A Seaway and J.W Jewett. "Physics for Scientists and engineers," Golden sunburst series, 8th Edition, 2010.
4. Sears, Zemansky & Young, "University Physics", 8th edition Addison-Wesley, USA, 2000.
- 5.

Calculus-I
MATH-101

Program-Discipline	BS – Mathematics
Credit Hours	4
Duration	18 Weeks

Course Title: Calculus-I**Course Code:** MATH-101**Course Description**

Calculus-I is an introductory course designed to provide students with a foundational understanding of differential and integral calculus. This course is essential for students pursuing degrees in mathematics, engineering, physical sciences, economics, and related fields.

Learning Outcomes

By the end of the course, students should be able to:

- Understand and apply the concept of a limit and continuity.
- Compute derivatives of various types of functions and apply them to solve real-world problems.
- Utilize derivatives for curve sketching and optimization problems.
- Understand and apply the Fundamental Theorem of Calculus.
- Compute definite and indefinite integrals and apply integration techniques to practical problems.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	Limits and Continuity: Rates of Change and Limits, Rules for Finding Limits, Target Values and Formal, Definitions of Limits
2	Extensions of the Limit Concept, Continuity, Tangent Lines
3	Derivatives: Derivative of a Function, Differentiation Rules,
4	Rates of Change, Derivatives of Trigonometric Functions, Chain Rule,
5	Implicit Differentiation and Rational Exponents, Related Rate of Change
6	Extreme Values of Functions, Mean Value Theorem, First Derivative Test for Local Extreme Values,
7	Graphing with y' and y'', Limits as $x \rightarrow \pm\infty$. Asymptotes, and Dominant Terms, Optimization, Linearization and Differentials, Newton's Method
8	REVISION
9	MID TERM EXAMS
10	Integration: Indefinite Integrals,
11	Differential Equations, Initial Value Problems, and Mathematical Modeling,
12	Integration by Substitution-Running the Chain Rule Backward,
13	Estimating with Finite Sums, Riemann Sums and Definite Integrals, Properties,
14	Substitution in Definite Integrals, Numerical Integration
15	Areas between Curves, Finding Volumes by Slicing, Volumes of Solids of Revolution-
16	- Disks and Washers, Cylindrical Shells, Lengths of Plane Curves, Areas of Surfaces of Revolution, Moments and - Centers of Mass, Work and other Modeling Applications.
17	REVISION
18	END SEMESTER EXAMS

Recommended Books

1. Calculus by G.B Thomas, 14th edition 2018.
2. Calculus with Analytic Geometry by Ron Larson, 2002
3. Calculus with Analytic Geometry by George Simon, 1996
4. Introduction to Calculus and Analytic Geometry by Gillett, 2008

Semester-II Course Outlines

Expository Writing
GEN-104

Program-Discipline	BS – Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Expository Writing**Course Code:** GEN-104**Course Description**

This required course focuses on writing clearly and cogently, overcoming prevalent errors in language, mastering rules in grammar and syntax, and developing skills in organizing and structuring different forms of writing. The course consists of two major parts: the reading section focuses on recognizing a topic sentence, skimming, scanning, use of cohesive devices, identifying facts and opinions, guess meanings of unfamiliar words etc. The writing section deals with the knowledge and the use of various grammatical components and essay writing skills which would enable students to write effective and advanced expository writing.

Course Learning Outcomes (CLOs)

Upon successful completion of the course, the student will be able to:

CLO No.	CLO Description
CLO-1	Develop their writing through a series of steps, the first of which is critical reading of passages.
CLO-2	Practice writing to refine their macro level as well as micro level compositional skills for professional purposes.
CLO-3	Employ various modes like argument, cause and effect, narration, description, comparison and contrast exemplification and classification in their writings.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3×16) 48 contact hours in the semester. In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	- Understanding Expository Writing & its purpose - Differentiating expository writing from other forms of writing (descriptive, narrative, persuasive) - Basic features (It informs, it is factual, presented in a logical format, clear about its purpose; it is not subjective, not an attempt to change reader's mind or shape their perspective)
2	Reading for writing - Reading strategies - Critical Reading skills - Find specific and general information quickly - Distinguish between relevant and irrelevant information according to purpose for reading - Distinguish between fact and opinion
3	Writing Process Generating ideas (collecting information in various forms; brainstorming, mind maps, tables, lists, charts, outlining, clustering)
4	Ordering Information/Text organization - Chronology - Stages of a process - From general to specific and vice versa - From most important to least important - Advantages and disadvantages - Cause and effect - Comparison and contrast - Problem solution pattern
5	Drafting - Free Writing - Revisions - Editing - Paraphrasing
6	Coherence and Cohesion Cohesive devices (linking words, connectors, discourse

	markers, transitional words) • Paragraph Unity (Developing a single idea thoroughly and linking it to the rest of essay/writing) • Ordering sentences in a logical manner • Transition links in sentences
7	Ethical Considerations in Expository Writing • Honesty, integrity, and accuracy in writing • Avoiding bias and presenting balanced viewpoints • Respecting copy and intellectual property rights
8	• Paragraph writing: simple, compound and complex sentence structure • Practice with every day topics
9	MID TERM EXAMS
10	• Characteristics of narrative, descriptive, and argumentative paragraphs • Essay writing Descriptive, narrative, discursive, and argumentative
11	• Identification of tone, diction, voice
12	Writing for Professional Purposes • Cover Letter formats • Cover Letter writing • Resume formats • Resume Writing • Email format • Email etiquettes
13	Visual Communication in Expository Writing • Selecting appropriate images and visual elements • Creating effective charts, graphs, and infographics • Ensuring accessibility of visual content • Ensuring readability of visual content
14	Understanding AI Writing Context • Exploring the intersection of artificial intelligence and expository writing • Analyzing the role of AI in generating written content

15	- Identifying opportunities and challenges in AI-assisted writing - Ethical considerations in AI-generated writing
16	Writing for Digital Platforms - Adapting expository writing for online audiences - Blogging and content creation - Search engine optimization (SEO) for online writing - Engaging readers through social media and interactive blogs
17	REVISION
18	END TERM EXAMS

Recommended Books

1. On writing well: The classic guide to writing nonfiction, Zinsser W. Harper Perennial, 2006
2. A Practical Guide to Business Writing: Writing in English for Non-Native Speakers, Khaled Al- Maskari, Wiley, 2012
3. College Writing Skills with Readings, by John Langan, McGraw-Hill, 5th Edition.
4. High School English Grammar & Composition by Wren & Martin
5. Practical English Grammar by A. J. Thomson and A. V. Martinet

Quantitative Reasoning-II
GEN-105

Program-Discipline	BS – Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Quantitative Reasoning-II**Course Code:** GEN-105**Course Description**

To provide students with an understanding of the basic Mathematical and Financial Techniques used in Business Enterprises or Accountancy Profession through application of techniques rather than its theoretical aspect. To equip students with Mathematical Skills and Knowledge that can be applied to solve Financial Problems in field of Accountancy

Learning Outcome

After taking a course in Quantitative Reasoning II, students will be able to:

- Problem Solving: Categorize an unfamiliar problem; justify the categorization; and identify valid and reliable steps needed to address the problem.
- The topics for the Quantitative Reasoning course are:
- Number, Ratio, and Proportional Reasoning, Modeling, Matrix, algebra, linear equation, Interest
- Probability
- Statistics

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	Ratio, types of ratio and related problems
2	Proportion, types of proportion, problems related to different types of proportion.
3	Percentage, real world problems related to profit and loss and zakat calculation.
4	Quadratic equation in one variable and its applications. Solution of quadratic equations by prime factorization and completing squares.
5	Roots of quadratic equations and its types, relation between roots and coefficients of quadratic equation, formation of a quadratic equation when roots are given.
6	Format of a quadratic equation when roots are given, nature of roots of quadratic equation
7	Concept of perimeter and its application. perimeter of rectangle, square, triangle and circle.
8	Concept of area, problems involving area of rectangle, circle, cone, triangle square and cylinder
9	MID TERM EXAMS
10	Inequalities and their properties, solution of inequalities.
11	Absolute values, one variable equations involving absolute values and their solution.
12	Sequence and series, general term of a sequence and related problems
13	Arithmetic sequence and its general term with related problems.

14	• Arithmetic series, problems involving arithmetic series.
15	• Geometric sequence and its general term with related problems.
16	• Geometric series, finite and infinite geometric series, problems involving geometric series.
17	• Harmonic sequence its general term with related problems
18	END SEMESTER EXAMS

Recommended Books

1. Frank S. Budnick, "Applied Mathematics for Business, Economics, and Social Sciences", Latest Edition, McGraw-Hill Inc.
2. Charles D. Miller, Stanley A. Salzman, and Gary Clendenen, "Business Mathematics", Latest Edition, Addison-Wesley
3. Glencoe "Mathematics with Business Applications", Student Latest Edition, McGraw-Hill.

Islamic Studies
GEN-106

Program-Discipline	BS – Mathematics
Credit Hours	2
Duration	18 Weeks

Course Title: *Islamic Studies

Course Code: GEN-106

Course Description

This course provides Basic information about Islamic Studies to enhance understanding of the students regarding Islamic Civilization. This course discusses History of Islam, understanding of the worship and its usefulness, the basic concept of Quran Pak: wisdom, patience, loyalty. The course also discusses comparative analysis of Islam with other religions. The Concept and Value of Haqooq ul Ibad (Bandon Kay Haqooq) in Islam is discussed. The rights of people in Islamic Point of View are also discussed.

Course Learning Outcomes (CLOs)

CLO No.	CLO Description
CLO-1	<u>Understand</u> the basic concept of Islam and Quran Pak. To enhance the skill of the students for understanding of issues related to Faith and Religious Life.
CLO-2	<u>Understand</u> the concept of Haqooq ul Ibad in the light of Quran. <u>Develop</u> an understanding of Divine Verses, Mandates and Provisions concerning permissible Methods and Modes of Islamic Trade and Commerce.
CLO-3	<u>Enhance</u> the knowledge of Islam.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

تدریسی اوقات	تدریسی مواد
پہلا ہفتہ	• سورة البقرة: آیت نمبر 1 تا 5 اور آیت نمبر 284 تا 286، حدیث نمبر 1 ، 2
دوسرا ہفتہ	• سورة الاحزاب: آیت نمبر 6، 21، 32، 33، حدیث نمبر 3، 4
تیسرا ہفتہ	• سورة الاحزاب: آیت نمبر 40 اور آیت نمبر 56 تا 59، حدیث نمبر 5، 6
چوتھا ہفتہ	• سورة الفتح : آیت نمبر 29، سورة الصف: آیت نمبر 1 تا 9
پانچواں ہفتہ	• سورة الصف : آیت نمبر 10 تا 14، سورة الحجرات : آیت نمبر 1 تا 10
چھٹا ہفتہ	• سورة الحجرات : آیت نمبر 11 تا 18، حدیث نمبر 7، 8
ساتواں ہفتہ	• مطالعہ سیرت کی ضرورت اور اہمیت، نبی کریم ﷺ کی حکمت انقلاب: ہجرت، مؤاخات، میثاق مدینہ
آٹھواں ہفتہ	• اسلامی تہذیب کے خصائص، اسلامی تہذیب و ثقافت کے عالمی اثرات
نواں ہفتہ	وسط مدتی امتحانات
دسواں ہفتہ	• سورة الانعام: آیات نمبر 151 تا 153، حدیث نمبر 9، 10
گیارہواں ہفتہ	• سورة الفرقان: آیت نمبر 63 تا 77، سورة النحل: آیت نمبر 12 تا 14
بارہواں ہفتہ	• صلح حدیبیہ، خطبہ حجة الوداع، حدیث نمبر 11، 12
تیرہواں ہفتہ	• تشکیل اجتماعی و معاشرت اور اسوہ حسنہ، حدیث نمبر 13، 14، 15، 16
چودھواں ہفتہ	• مغربی تہذیب و ثقافت اور اسلام، مغربی تہذیب و ثقافت کے خصائص و اثرات
پندرہواں ہفتہ	• تہذیبوں کے تصادم کے نظریے کا تنقیدی جائزہ، حدیث نمبر 17، 18، 19، 20
سولہواں ہفتہ	• حدیث نمبر 21 تا 23، عشرہ مبشرہ • اسلامی معاشی نظام کے بنیادی تصورات • تجارت اور تجارت کے اسلامی طریقے • سورة البقرة: آیت نمبر 275 • سورة النساء: آیت نمبر 29 • حدیث نمبر 11، 12، 16
سترہواں ہفتہ	• حدیث نمبر 24 تا 26، امہات المؤمنین، اولاد النبی ﷺ
اٹھارہواں ہفتہ	اختتامی سمسٹر امتحانات

نوٹ: نصاب میں موجود منتخب آیات (از سورة البقرة، سورة الاحزاب، سورة الفتح، سورة الصف، سورة الحجرات، سورة الانعام، سورة الفرقان، سورة النحل) اور منتخب احادیث کا ترجمہ اور تشریح دونوں مطلوب ہیں۔

ضمیمہ: منتخب احادیثِ مبارکہ

1. عَنْ عُمَرَ بْنِ الْخَطَّابِ رَضِيَ اللَّهُ عَنْهُ قَالَ: سَمِعْتُ رَسُولَ اللَّهِ ﷺ يَقُولُ: إِنَّمَا الْأَعْمَالُ بِالنِّيَّاتِ، وَإِنَّمَا لِامْرِئٍ مَّا نَوَى، فَمَنْ كَانَتْ هِجْرَتُهُ إِلَى اللَّهِ وَرَسُولِهِ، فَهَاجَرَتْهُ إِلَى اللَّهِ وَرَسُولِهِ، وَمَنْ كَانَتْ هِجْرَتُهُ إِلَى دُنْيَا يُصِيبُهَا أَوْ امْرَأَةٍ يَتَرَوُّجُهَا، فَهَاجَرَتْهُ إِلَى مَا هَاجَرَ إِلَيْهِ. صحيح البخاري (8 / 140)
2. عَنْ عُثْمَانَ رَضِيَ اللَّهُ عَنْهُ، عَنِ النَّبِيِّ ﷺ قَالَ: خَيْرُكُمْ مَنْ تَعَلَّمَ الْقُرْآنَ وَعَلَّمَهُ. صحيح البخاري (6 / 192)
3. عَنْ مَالِكِ بْنِ أَنَسٍ قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ تَرَكْتُ فِيكُمْ أَمْرَيْنِ لَنْ تَضِلُّوا مَا تَمَسَّكْتُمْ بِهِمَا: كِتَابَ اللَّهِ وَسُنَّةَ رَسُولِهِ. موطأ مالك (5 / 1323)
4. عَنِ ابْنِ عُمَرَ رَضِيَ اللَّهُ عَنْهُمَا قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ: بُنِيَ الْإِسْلَامُ عَلَى خَمْسٍ، شَهَادَةِ أَنْ لَا إِلَهَ إِلَّا اللَّهُ، وَأَنَّ مُحَمَّدًا عَبْدُهُ وَرَسُولُهُ، وَإِقَامِ الصَّلَاةِ، وَإِيتَاءِ الزَّكَاةِ، وَحَجِّ الْبَيْتِ، وَصَوْمِ رَمَضَانَ. صحيح مسلم (1 / 45)
5. عَنْ عُمَرَ بْنِ الْخَطَّابِ رَضِيَ اللَّهُ عَنْهُ قَالَ: بَيْنَمَا نَحْنُ عِنْدَ رَسُولِ اللَّهِ ﷺ ذَاتَ يَوْمٍ، إِذْ طَلَعَ عَلَيْنَا رَجُلٌ شَدِيدُ بَيَاضِ الثِّيَابِ، شَدِيدُ سَوَادِ الشَّعْرِ، لَا يُرَى عَلَيْهِ أَثَرُ السَّفَرِ، وَلَا يَعْرِفُهُ مِنَّا أَحَدٌ، حَتَّى جَلَسَ إِلَى النَّبِيِّ ﷺ، فَأَسْنَدَ رُكْبَتَيْهِ إِلَى رَسُولِ اللَّهِ ﷺ: «الْإِسْلَامُ أَنْ تَشْهَدَ أَنْ لَا إِلَهَ إِلَّا اللَّهُ وَأَنَّ مُحَمَّدًا رَسُولُ اللَّهِ ﷺ، وَتُقِيمَ الصَّلَاةَ، وَتُؤْتِيَ الزَّكَاةَ، وَتَصُومَ رَمَضَانَ، وَتَحُجَّ الْبَيْتَ إِنْ اسْتَطَعْتَ إِلَيْهِ سَبِيلًا.» قَالَ: صَدَقْتَ. قَالَ: فَعَجِبْنَا لَهُ يَسْأَلُهُ وَيُصَدِّقُهُ. قَالَ: فَأَخْبِرْنِي عَنِ الْإِيمَانِ. قَالَ: «أَنْ تُؤْمِنَ بِاللَّهِ، وَمَلَائِكَتِهِ، وَكُتُبِهِ، وَرُسُلِهِ، وَالْيَوْمِ الْآخِرِ، وَتُؤْمِنَ بِالْقَدَرِ خَيْرِهِ وَشَرِّهِ» قَالَ: صَدَقْتَ. قَالَ: فَأَخْبِرْنِي عَنِ الْإِحْسَانِ. قَالَ: «أَنْ تَعْبُدَ اللَّهَ كَأَنَّكَ تَرَاهُ، فَإِنْ لَمْ تَكُنْ تَرَاهُ فَإِنَّهُ يَرَاكَ» قَالَ: فَأَخْبِرْنِي عَنِ السَّائِلِ. قَالَ: «أَنْ تَلِدَ الْأُمَّةَ رَبَّتْهَا، وَأَنْ تَرَى الْخُفَاءَ الْعُزْرَةَ الْعَالَةَ رِعَاءَ الشَّاءِ يَتَطَاوَلُونَ فِي الْبُنْيَانِ» قَالَ: ثُمَّ انْطَلَقَ فَلِيَتْهُ مَلِيًّا، ثُمَّ قَالَ لِي: «يَا عُمَرُ! أَتَدْرِي مَنْ السَّائِلُ؟» قُلْتُ: اللَّهُ وَرَسُولُهُ أَعْلَمُ. قَالَ: «فَإِنَّهُ جِبْرِيلُ أَتَاكُمْ يُعَلِّمُكُمْ دِينَكُمْ.» (صحيح مسلم: 1 / 37)

6. عَنْ سَبْرَةَ بْنِ مَعْبِدٍ الْجُهَنِيِّ رَضِيَ اللَّهُ عَنْهُ قَالَ: قَالَ النَّبِيُّ ﷺ: مُرُوا الصَّيِّ بِالصَّلَاةِ إِذَا بَلَغَ سَبْعَ سِنِينَ وَإِذَا بَلَغَ عَشَرَ سِنِينَ فَاضْرِبُوهُ عَلَيْهَا. (سنن أبي داود (1/ 133) حسن) وَاخْرَجَهُ التِّرْمِذِيُّ فِي سَنَنِهِ وَلَفْظُهُ: عَلِّمُوا الصَّيِّ الصَّلَاةَ ابْنَ سَبْعَ سِنِينَ، وَاضْرِبُوهُ عَلَيْهَا ابْنَ عَشْرِ. سنن الترمذي ت شاكر (2/ 259) حسن صحيح

7. عَنْ مُعَاوِيَةَ رَضِيَ اللَّهُ عَنْهُ قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ: مَنْ يُرِدِ اللَّهُ بِهِ خَيْرًا يُفَقِّهْهُ فِي الدِّينِ. صحيح البخاري (1/ 25)

8. عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ: «مَنْ سَلَكَ طَرِيقًا يَلْتَمِسُ فِيهِ عِلْمًا، سَهَّلَ اللَّهُ لَهُ بِهِ طَرِيقًا إِلَى الْجَنَّةِ، وَمَا اجْتَمَعَ قَوْمٌ فِي بَيْتٍ مِنْ بُيُوتِ اللَّهِ، يَتْلُونَ كِتَابَ اللَّهِ، وَيَتَذَكَّرُونَ بَيْنَهُمْ، إِلَّا نَزَلَتْ عَلَيْهِمُ السَّكِينَةُ، وَغَشِيَتْهُمْ الرَّحْمَةُ وَخَفَّتْهُمْ الْمَلَائِكَةُ، وَذَكَرَهُمُ اللَّهُ فِيمَنْ عِنْدَهُ، وَمَنْ بَطَأَ بِهِ عَمَلُهُ، لَمْ يُسْرِعْ بِهِ نَسَبُهُ». صحيح مسلم (4/ 2074)

9. عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ قَالَ: كَانَ رَسُولُ اللَّهِ ﷺ يَقُولُ: اَللّهُمَّ إِنِّي أَعُوذُ بِكَ مِنَ الْأَرْبَعِ: مِنْ عِلْمٍ لَا يَنْفَعُ، وَمِنْ قَلْبٍ لَا يَخْشَعُ، وَمِنْ نَفْسٍ لَا تَشْبَعُ، وَمِنْ دُعَاءٍ لَا يُسْمَعُ. مسند أحمد ط الرسالة (14/ 188) سنن أبي داود (2/ 92) سنن ابن ماجه (2/ 1261)

10. عَنْ ابْنِ مَسْعُودٍ رَضِيَ اللَّهُ عَنْهُ عَنِ النَّبِيِّ ﷺ قَالَ: «لَا تَزُولُ قَدَمُ ابْنِ آدَمَ يَوْمَ الْقِيَامَةِ مِنْ عِنْدِ رَبِّهِ حَتَّى يُسْأَلَ عَنْ خَمْسٍ: عَنْ عُمْرِهِ فِيمَ أَفْنَاهُ، وَعَنْ شَبَابِهِ فِيمَ أَبْلَاهُ، وَمَالِهِ مِنْ أَيْنَ اكْتَسَبَهُ وَفِيمَ أَنْفَقَهُ، وَمَاذَا عَمِلَ فِيمَا عَلِمَ. سنن الترمذي ت شاكر (4/ 612)

11. عَنْ عَبْدِ اللَّهِ رَضِيَ اللَّهُ عَنْهُ قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ: طَلَبُ كَسْبِ الْخَلَالِ فَرِيضَةٌ بَعْدَ الْفَرِيضَةِ. شعب الإيمان (11/ 176)

12. عَنْ أَبِي سَعِيدٍ رَضِيَ اللَّهُ عَنْهُ عَنِ النَّبِيِّ ﷺ قَالَ: اَلتَّاجِرُ الصُّدُوقُ الْأَمِينُ مَعَ النَّبِيِّينَ وَالصِّدِّيقِينَ وَالشُّهَدَاءِ. سنن الدارمي (3/ 1653) سنن الترمذي ت شاكر (3/ 507)

13. عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ أَنَّ رَسُولَ اللَّهِ ﷺ قَالَ: أَتَذَرُونَ مَا الْمُفْلِسُ؟ قَالُوا: الْمُفْلِسُ فِينَا مَنْ لَا دِرْهَمَ لَهُ وَلَا مَتَاعَ. فَقَالَ: إِنَّ الْمُفْلِسَ مِنْ أُمَّتِي يَأْتِي يَوْمَ الْقِيَامَةِ بِصَلَاةٍ، وَصِيَامٍ، وَزَكَاةٍ، وَيَأْتِي قَدْ شَتَمَ هَذَا، وَقَذَفَ هَذَا، وَأَكَلَ مَالَ هَذَا، وَسَقَكَ دَمَ هَذَا، وَضَرَبَ هَذَا، فَيُعْطَى هَذَا مِنْ حَسَنَاتِهِ، وَهَذَا مِنْ حَسَنَاتِهِ، فَإِنْ فَنِيَتْ حَسَنَاتُهُ قَبْلَ أَنْ يُقْضَى مَا عَلَيْهِ، أُخِذَ مِنْ خَطَايَاهُمْ فَطُرِحَتْ عَلَيْهِ، ثُمَّ طُرِحَ فِي النَّارِ. صحيح مسلم (4/ 1997)

14. عَنْ أَبِي الدَّرْدَاءِ رَضِيَ اللَّهُ عَنْهُ عَنِ النَّبِيِّ ﷺ قَالَ: إِنَّ أَثْقَلَ شَيْءٍ يُوَضَّعُ فِي مِيزَانِ الْمُؤْمِنِينَ يَوْمَ الْقِيَامَةِ خُلُقٌ حَسَنٌ. وَإِنَّ اللَّهَ يُبْغِضُ الْفَاحِشَ الْبَذِيَّ. شرح السنة للبغوي (13 / 78)

15. عَنْ ابْنِ عَبَّاسٍ رَضِيَ اللَّهُ عَنْهُمَا أَنَّ رَسُولَ اللَّهِ ﷺ قَالَ: أَرْبَعُ مَنْ أُعْطِيَهُنَّ فَقَدْ أُعْطِيَ خَيْرَ الدُّنْيَا وَالْآخِرَةِ؛ قَلْبًا شَاكِرًا، وَلِسَانًا ذَاكِرًا، وَبَدَنًا عَلَى الْبَلَاءِ صَابِرًا، وَزَوْجَةً لَا تَبْغِيهِ خَوْنًا فِي نَفْسِهَا وَلَا مَالِهِ. المعجم الأوسط (7 / 179)

16. عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ، عَنِ النَّبِيِّ ﷺ قَالَ: اجْتَنِبُوا السَّبْعَ الْمُؤَيَّقَاتِ. قَالُوا: يَا رَسُولَ اللَّهِ! وَمَا هُنَّ؟ قَالَ: الشِّرْكُ بِاللَّهِ، وَالسَّحَرُ، وَقَتْلُ النَّفْسِ الَّتِي حَرَّمَ اللَّهُ إِلَّا بِالْحَقِّ، وَأَكْلُ الرِّبَا، وَأَكْلُ مَالِ الْيَتِيمِ، وَالتَّوَلَّى يَوْمَ الرَّخْفِ، وَقَذْفُ الْمُحْصَنَاتِ الْمُؤْمِنَاتِ الْغَافِلَاتِ. صحيح البخاري (4 / 10)

17. عَنْ أَبِي سَعِيدٍ الْخُدْرِيِّ رَضِيَ اللَّهُ عَنْهُ عَنِ النَّبِيِّ ﷺ قَالَ: مَنْ رَأَى مِنْكُمْ مُنْكَرًا فَلْيُغَيِّرْهُ بِيَدِهِ، فَإِنْ لَمْ يَسْتَطِعْ فَبِلِسَانِهِ، فَإِنْ لَمْ يَسْتَطِعْ فَبِقَلْبِهِ، وَذَلِكَ أَضْعَفُ الْإِيمَانِ. صحيح مسلم (1 / 69)

18. يُجَاءُ بِالرَّجُلِ يَوْمَ الْقِيَامَةِ، فَيُلْقَى فِي النَّارِ، فَتَنْدَلِقَ أَفْتَابُهُ فِي النَّارِ، فَيَطْخُنُ فِيهَا كَطَخْنِ الْجِمَارِ بِرَحَاهُ، فَيَجْتَمِعُ أَهْلُ النَّارِ عَلَيْهِ، فَيَقُولُونَ: أَيُّ فُلَانٍ! مَا شَأْنُكَ؟ أَلَيْسَ كُنْتَ تَأْمُرُنَا بِالْمَعْرُوفِ، وَتَنْهَانَا عَنِ الْمُنْكَرِ؟ قَالَ: كُنْتُ آمُرُكُمْ بِالْمَعْرُوفِ وَلَا آتِيهِ، وَأَنْهَأَكُمْ عَنِ الْمُنْكَرِ وَآتِيهِ. شرح السنة للبغوي (14 / 352، 351)

19. عَنْ أَنَسٍ رَضِيَ اللَّهُ عَنْهُ، عَنِ النَّبِيِّ ﷺ قَالَ: وَالَّذِي نَفْسِي بِيَدِهِ، لَا يُؤْمِنُ عَبْدٌ حَتَّى يُحِبَّ لِأَخِيهِ مَا يُحِبُّ لِنَفْسِهِ. صحيح مسلم (1 / 68)

20. عَنِ النُّعْمَانِ بْنِ بَشِيرٍ رَضِيَ اللَّهُ عَنْهُ يَقُولُ: قَالَ رَسُولُ اللَّهِ ﷺ: تَرَى الْمُؤْمِنِينَ فِي تَرَاحِمِهِمْ وَتَوَادِّهِمْ وَتَعَاطُفِهِمْ، كَمَثَلِ الْجَسَدِ، إِذَا اشْتَكَى عُضْوًا تَدَاعَى لَهُ سَائِرُ جَسَدِهِ بِالسَّهْرِ وَالْحُمَى. صحيح البخاري (8 / 10)

21. عَنْ عَبْدِ اللَّهِ بْنِ عُمَرَ رَضِيَ اللَّهُ عَنْهُمَا: أَنَّ رَسُولَ اللَّهِ ﷺ قَالَ: أَلَا كُلُّكُمْ رَاعٍ وَكُلُّكُمْ مَسْئُولٌ عَنْ رَعِيَّتِهِ، فَالْإِمَامُ الَّذِي عَلَى النَّاسِ رَاعٍ وَهُوَ مَسْئُولٌ عَنْ رَعِيَّتِهِ، وَالرَّجُلُ رَاعٍ عَلَى أَهْلِ بَيْتِهِ، وَهُوَ مَسْئُولٌ عَنْ رَعِيَّتِهِ، وَالْمَرْأَةُ رَاعِيَةٌ عَلَى أَهْلِ بَيْتِ زَوْجِهَا، وَوَلَدِهِ وَهِيَ مَسْئُولَةٌ عَنْهُمْ، وَعَبْدُ الرَّجُلِ رَاعٍ عَلَى مَالِ سَيِّدِهِ وَهُوَ مَسْئُولٌ عَنْهُ، أَلَا فَكُلُّكُمْ رَاعٍ وَكُلُّكُمْ مَسْئُولٌ عَنْ رَعِيَّتِهِ. صحيح البخاري (9 / 62)

22. عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ، قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ: «مَثَلِي وَمَثَلُ الْأَنْبِيَاءِ كَمَثَلِ قَمَرٍ أَحْسَنَ بُنْيَانُهُ، ثَرَكٌ مِنْهُ مَوْضِعُ لَيْتَةٍ، فَطَافَ بِهَا النَّظَّارُ يَتَعَجَّبُونَ مِنْ حُسْنِ بِنَائِهِ إِلَّا مَوْضِعَ تِلْكَ اللَّيْتَةِ، لَا يَعِيبُونَ سِوَاهَا، فَكُنْتُ أَنَا

سَدَدَتْ مَوْضِعَ تِلْكَ اللَّيْنَةِ، خُتِمَ بِي الْبُنْيَانُ، وَخُتِمَ بِي الرُّسْلُ» وفي رواية: فَأَنَا اللَّيْنَةُ، وَأَنَا خَاتَمُ النَّبِيِّينَ. شرح السنة للبغوي (13/ 201)

23. عَنْ أَنَسِ بْنِ مَالِكٍ رَضِيَ اللَّهُ عَنْهُ، عَنِ النَّبِيِّ ﷺ قَالَ: " أَرْحَمُ أُمَّتِي بِأُمَّتِي أَبُو بَكْرٍ، وَأَشَدُّهُمْ فِي أَمْرِ اللَّهِ عُمَرُ وَأَصْدَقُهُمْ حَيَاءٌ عُثْمَانُ وَأَقْضَاهُمْ عَلِيٌّ، وَأَفْرَضُهُمْ زَيْدُ بْنُ ثَابِتٍ، وَأَقْرَوُهُمْ لِكِتَابِ اللَّهِ أَبِي بْنُ كَعْبٍ، وَأَعْلَمُهُمْ بِالْخَلَالِ وَالْحَرَامِ مُعَاذُ بْنُ جَبَلٍ، أَلَا وَإِنَّ لِكُلِّ أُمَّةٍ أَمِيئًا، وَإِنَّ أَمِينَ هَذِهِ الْأُمَّةِ أَبُو عُبَيْدَةَ بْنُ الْجَرَّاحِ. مسند أحمد ط الرسالة (21/ 406) سنن ابن ماجه (1/ 55)

24. عَنْ أَبِي بَكْرَةَ رَضِيَ اللَّهُ عَنْهُ يَقُولُ: رَأَيْتُ رَسُولَ اللَّهِ ﷺ عَلَى الْمِنْبَرِ وَالْحَسَنُ بْنُ عَلِيٍّ إِلَى جَنْبِهِ، وَهُوَ يُقِيلُ عَلَى النَّاسِ مَرَّةً، وَعَلَيْهِ أُخْرَى وَيَقُولُ: «إِنَّ ابْنِي هَذَا سَيِّدٌ وَلَعَلَّ اللَّهَ أَنْ يُصْلِحَ بِهِ بَيْنَ فِئَتَيْنِ عَظِيمَتَيْنِ مِنَ الْمُسْلِمِينَ. صحيح البخاري (3/ 186)

25. عَنْ عُمَرَ بْنِ حُصَيْنٍ رَضِيَ اللَّهُ عَنْهُمَا، يَقُولُ: قَالَ رَسُولُ اللَّهِ ﷺ: " خَيْرُ أُمَّتِي قَرْنِي، ثُمَّ الَّذِينَ يَلُونَهُمْ، ثُمَّ الَّذِينَ يَلُونَهُمْ. صحيح البخاري (5/ 2)

26. عَنْ جَابِرِ بْنِ عَبْدِ اللَّهِ رَضِيَ اللَّهُ عَنْهُ قَالَ: خَطَبَنَا رَسُولُ اللَّهِ ﷺ فِي وَسْطِ أَيَّامِ التَّشْرِيقِ خُطْبَةً الْوَدَاعِ، فَقَالَ: " يَا أَيُّهَا النَّاسُ، إِنَّ رَبَّكُمْ وَاحِدٌ، وَإِنَّ أَبَاكُمْ وَاحِدٌ، أَلَا لَا فَضْلَ لِعَرَبِيٍّ عَلَى عَجَمِيٍّ، وَلَا لِعَجَمِيٍّ عَلَى عَرَبِيٍّ، وَلَا لِأَحْمَرَ عَلَى أَسْوَدَ، وَلَا أَسْوَدَ عَلَى أَحْمَرَ، إِلَّا بِالتَّقْوَى، إِنَّ أَكْرَمَكُمْ عِنْدَ اللَّهِ أَتَقَاكُمْ، أَلَا هَلْ بَلَّغْتُ؟ "، قَالُوا: بَلَى يَا رَسُولَ اللَّهِ، قَالَ: " فَلْيُبَلِّغِ الشَّاهِدُ الْغَائِبَ. شعب الإيمان (7/ 132)

Recommended Books

1. Translation of the Holy Qur'an by Abdullah Yousuf Ali
2. The Glorious Qur'an by Dr Muhammad Tahir-ul-Qadri
3. Hameed Ullah Muhammad, "Muslim Conduct of State"
4. Hameed Ullah Muhammad, 'Introduction to Islam'
5. Hussain Hamid Hassan, "An Introduction to the Study of Islamic Law"
Leaf Publication Islamabad, Pakistan.
6. Ahmad Hasan, "Principles of Islamic Jurisprudence" Islamic Research
Institute, International Islamic University, Islamabad (Latest Edition)
7. Mir Waliullah, "Muslim Jurisprudence and the Quranic Law of Crimes"

Islamic Book Service (Latest Edition)

8. H.S. Bhatia, "Studies in Islamic Law, Religion and Society" Deep & Deep Publications New Delhi (Latest Edition)
9. Dr. Muhammad Zia-ul-Haq, "Introduction to Al Sharia Al Islamia" Allama Iqbal Open University, Islamabad (Latest Edition)

نصابی کتاب

تفہیم اسلامیات از پروفیسر محمد نواز چوہدری و دیگر یا کوئی بھی کتاب جس میں کورس آؤٹ لائن کے مطابق مواد موجود ہو

حوالہ جاتی کتاب

تفسیر القرآن العظیم از ابن کثیر
تفسیر روح المعانی از شہاب الدین محمود آلوسی
معارف القرآن از مفتی محمد شفیع
ضیاء القرآن از پیر کرم شاہ الازہری
مشکاۃ المصابیح از منظور تبریزی
معارف الحدیث از منظور نعمانی
سیرت النبی ﷺ از شبلی نعمانی و سید سلمان ندوی
الرحیق المختوم از صفی الرحمن مبارکپوری
سیر الصحابہ از شاہ معین الدین ندوی
تاریخ اسلام از شاہ معین الدین ندوی
عشرہ مبشرہ از محمود احمد ظفر
اسلامی تہذیب اور اس کے اصول و مبادی از سید ابوالاعلیٰ مودودی
اسلام اور جدید ذہن کے شبہات از سید محمد قطب
خطبات بہاولپور از ڈاکٹر محمد حمید اللہ

Other Reference Materials

1. Tafheem e Islamiyat by Prof. Dr. M. Nawaz Chohdary
2. Islamic Education by M D Zafar
3. Islamic Studies by Rafique Mohammad

Statistics
STAT-111

Program-Discipline	BS – Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Statistics**Course Code:** STAT-111**Course Description**

This course deals with statistical concepts, such as data presentation, graphs, central tendency and dispersion measures, correlation, and index numbers.

Course Objectives

This course helps to equip and understand descriptive statistical analysis to a problem for the students. The aims of course are to:

- Understand techniques of data collection,
- Give tools for data presentation including classification, organization, and graphical presentation of data,
- Analyze data by using measures of central tendency, dispersion, skewness, and kurtosis,
- Give a survey of basic probability concepts, and
- Write up the results of statistical analysis in a report for decision-makers.

Learning Outcomes

Upon successful completion of the course, the student will:

- Understand and be able to Calculate basic statistical concepts (tabular and graphic representation of data, measures of central tendency, dispersion and asymmetry, correlation).
- Apply knowledge to Solve simple tasks using computer software, such as MS Excel, SPSS, and R software.
- Based on the acquired knowledge to Interpret the meaning of the calculated statistical results

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3x16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	- Descriptive and inferential statistics, basic terminologies including variable and constant, population and sample, parameter and statistic. - Type of measurement scales: nominal, ordinal, interval, and ratio. - Types of data: univariate, bivariate, and multivariate data, primary and secondary data, quantitative and qualitative data, time series, cross-sectional and pooled data. - The concept of significant digits and rounding off numbers, errors, bias, and unbiased.
2	- Data collection methods. - Classification and tabulation, the frequency distribution, the cumulative frequency distribution, the relative frequency distribution, and the percentage frequency distribution.
3	- Graphic and diagrammatic representation: simple and multiple bar charts, pie charts, histograms, and frequency curves. - Types of averages for group and ungroup data including arithmetic mean, weighted mean, trimmed mean, median, and mode.
4	- Empirical relation between arithmetic mean, median, and mode. - Calculation of quartiles, deciles, and percentiles for group and ungroup data.
5	- Definition of outliers and their detection. - The summary measures and box plot. - Advantages and limitations of mean, median, and mode.
6	- Absolute and relative measure of dispersion. - Computation of range, mean deviation, variance, standard deviation, and coefficient of variation for grouped and ungrouped data.

7	• Definition and interpretation of variance and standard deviation, Properties, merits, and demerits of measure of dispersions.
8	• Skewness: Karl Pearson's coefficient of skewness, Bowley's coefficient of skewness, and kurtosis.
9	MID TERM EXAMS
10	• Introduction of Regression and Correlation • Cause and effect relationships, examples.
11	• Simple linear regression, estimation of parameters and their interpretation r and R • Correlation. Coefficient of linear correlation, its estimation and interpretation.
12	• Multiple regression and interpretation of its parameters. Examples Covariance and correlation analysis
13	• Introduction of Time Series Analysis • Components of Time Series • Depersonalization of Data
14	• Moving average method • Least square method. • Defining an Index Number • Un-weighted Aggregative Index
15	• Weighted Aggregative Index • Average of Relative Methods
16	• Quantity and Value Indices
17	• Lectures • In-Class Activities • Assignments • Practical (Basics of Excel / SPSS, optional)
18	END SEMESTER EXAMS

Recommended Books

1. Anderson, Sweeney and Williams, Statistics for Business and Economics, South-Western, Cengage Learning. (latest edition)
2. Weiss, N. A. Elementary Statistics, Pearson. (latest edition)
3. McClave, J. and Sincich, T. Statistics Pearson. (latest edition)
4. Gerald, K. Statistics for Management and Economics, Pearson. (latest edition)
5. Lind, Douglas A., Marshal, William G. and Mason, Robert D., (2015) Statistical Techniques in Business and Economics (16th edition). Boston: McGraw Hill, 2003.

Calculus-II
MATH-111

Program-Discipline	BS – Mathematics
Credit Hours	4
Duration	18 Weeks

Course Title: Calculus-II**Course Code:** MATH-111**Course Description**

Calculus-II is a continuation of Calculus I, designed to further develop students' understanding of integral calculus and introduce them to sequences, series, and other advanced topics. This course is crucial for students in mathematics, engineering, physical sciences, economics, and related fields.

Learning Outcomes

By the end of the course, students should be able to:

- Utilize advanced integration techniques to evaluate integrals.
- Apply integration to solve problems involving areas, volumes, arc lengths, and work.
- Understand and use parametric equations and polar coordinates.
- Analyze the convergence of sequences and series.
- Solve basic differential equations and apply them to real-world problems.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3×16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	Transcendental Functions: Inverse Functions and Their Derivatives, Natural Logarithms, Exponential Function, a^x and $\log_a^x$
2	Growth and Decay, L' Hospital's Rule, Relative Rates of Growth, Inverse Trigonometric Functions,
3	Derivatives of Inverse Trigonometric Functions; Integrals, Hyperbolic Functions, First Order Differential Equations, Euler's Numerical Method,
4	Techniques of Integration: Basic Integration Formulas, Integration by Parts
5	Partial Fractions, Trigonometric Substitutions,
6	Improper Integrals, Infinite Sequences, Limits of Sequences of Numbers, Theorems for Calculating Limits of Sequences
7	- Infinite Series, - Integral Test for Series of Nonnegative Terms, Comparison Tests for Series of Nonnegative Terms
8	REVISION
9	MID TERM EXAMS
10	Alternating Series, Absolute and Conditional Convergence,
11	Power Series, Taylor and Maclaurin Series, Convergence of Taylor Series; Error Estimates, Applications of Power Series
12	Conic Sections, Parameterized Curves, and Polar Coordinates: Conic Sections and Quadratic Equations, Classifying Conic Sections by Eccentricity,
13	Quadratic Equations and Rotations, Parameterizations of Plane Curves, Calculus with Parameterized Curves
14	Polar Coordinates, Graphing in Polar Coordinates, Polar Equations for Conic Sections, Integration on Polar Coordinates

15	• Vectors and Analytic Geometry in Space: Vectors in the plane, Cartesian (Rectangular) Coordinates and Vectors in Space
16	• Dot Products, Cross Products, Lines and Planes in Space, Cylinders and Quadric Surfaces, Cylinders and Spherical Coordinates
17	REVISION
18	END SEMESTER EXAMS

Recommended Books

1. Calculus by G.B Thomas, 14th edition 2018.
2. Calculus with Analytic Geometry by Ron Larson, 2002
3. Calculus with Analytic Geometry by George Simon, 1996
4. Introduction to Calculus and Analytic Geometry by Gillett, 2008

Elementary Set Theory
MATH-112

Program-Discipline	BS – Mathematics
Credit Hours	3
Duration	18 Weeks

Course Title: Elementary Set Theory**Course Code:** MATH-112**Course Description**

Elementary Set Theory is an introductory course designed to provide students with a comprehensive understanding of set theory, a crucial foundation for various areas of mathematics. The course covers the basic concepts, operations, and properties of sets, along with their applications in mathematical reasoning and problem-solving.

Learning Outcomes

By the end of the course, students should be able to:

- Understand and use the language of set theory to describe and analyze sets.
- Perform and interpret basic set operations and their properties.
- Understand the concepts of relations and functions and their applications.
- Compare the sizes of sets using cardinality.
- Apply set-theoretical concepts to solve problems in mathematics and related fields.

Weightage of Exams

Theory Subjects (May Include Practical / Lab Element)

Exam		Weightage
(a)	End-Semester Exam	50
(b)	Mid-Semester Exam	30
(c)	Quiz / assignment*	20
(d)	Practical / Lab / Projects	
(e)	Class Participation	
Total		100

*One Quiz and One Assignment per Semester is mandatory. More than One Quiz/Assignment may be taken and average or the best score shall be credited in such case.

One (01) Credit Hour of Theory is equivalent to One (01) contact hour per week i.e. One Hour teaching per Week. In the semester of 16 weeks of teaching, 3 Credit Hours of Theory Course means (3×16) 48 contact hours in the semester.

In the semester of 16 weeks of teaching, one (01) Credit Hour of Lab is equivalent to 3 contact hours per week for 16 weeks i.e. $3 \times 16 = 48$ contact hours in the semester.

Week Wise Course Outline

Week	Contents
1	- Cardinality - Equivalent sets, finite and infinite sets - Denumerable sets
2	Countable and uncountable sets
3	Cardinal numbers, addition and multiplication of cardinals, Cartesian product as sets of functions
4	Different types of infinity (Cantor's contribution)
5	- Partially ordered sets, Hasse diagrams - Totally ordered sets
6	Maximal and minimal elements
7	Upper and lower bound
8	REVISION
9	MID TERM EXAMS
10	Multiplication of ordinal numbers
11	Axiom of Choice
12	Zorn's lemma
13	Paradoxes in Set Theory
14	Cantor's paradox,
15	Russell's paradox and others.
16	Zorn's lemma
17	REVISION
18	END SEMESTER EXAMS

Recommended Books

1. A. A. Fraenkel, Abstract Set Theory, (North-Holland Publishing, Amsterdam, 1966).
2. Patrick Suppes, Axiomatic Set Theory, (Dover Publications, Inc., New York, 1972). P. R. Halmos, Naive Set Theory, (Van Nostrand, New York, 1960).
3. B. Rotman and G. T. Kneebone, The Theory of Sets and Transfinite Numbers, (Oldbourne, London, 1968)
4. Douglas Smith, Maurice Eggen and Richard St. Andre: A Transition to Advanced Mathematics, (Brooks/Cole, 2001)